Name: _____

WORKSHEET

Permutation calculations

1 In how many ways can each of the following numbers of different objects be placed in an ordered list?

a 4

b 7

c 3

d 10

e 8

f 6

2 Write the following in the form:

i $\dfrac{n!}{(n-r)!}$

ii $(n \times (n-1) \times (n-2) \times \ldots \times (n-r+1)$

a ${}^{11}P_4$

b ${}^{5}P_2$

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c ${}^{19}P_5$

d ${}^{10}P_7$

e ${}^{7}P_3$

f ${}^{9}P_8$

3 Calculate the following values.

a ${}^{6}P_2$

b ${}^{8}P_5$

c ${}^{5}P_3$

d ${}^{12}P_9$

e ${}^{10}P_6$

f ${}^{5}P_1$

4 Explain the difference between:

a ${}^4P_1$ and ${}^4P_4$.

b ${}^6P_3$ and ${}^9P_3$.

5 How many different ways can:

a a line of four people be chosen from seven people?

b a line of five people be chosen from ten people?

c an ordered display of six paintings be chosen from nine paintings?

d a tiered selection of three chocolates be chosen from eight different chocolates?

e a hierarchy of four people be chosen from twenty people?

f a winner and runner up be chosen from seven different contestants?

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Answers

1 **a** 24

b 5040

c 6

d 3 628 800

e 40 320

f 720

2 **a** **i** $\frac{11!}{(11-4)!}$ **ii** $11 \times 10 \times 9 \times 8$

b **i** $\frac{5!}{(5-2)!}$ **ii** 5×4

c **i** $\frac{19!}{(19-5)!}$ **ii** $19 \times 18 \times 17 \times 16$

d **i** $\frac{10!}{(10-7)!}$ **ii** $10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4$

e **i** $\frac{7!}{(7-3)!}$ **ii** $7 \times 6 \times 5$

f **i** $\frac{9!}{(9-8)!}$ **ii** $9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2$

3 **a** 30

b 6 720

c 60

d 79 833 600

e 151 200

f 5

4 **a** ${}^4P_1$ is the number of ways one object can be put into order from four objects (4), and ${}^4P_4$ is the number of ways four objects can be put into order from four objects (24).

b ${}^6P_3$ is the number of ways three objects can be put into order from six objects (120), and ${}^9P_3$ is the number of ways three objects can be put into order from nine objects (504).

5 **a** ${}^7P_4 = 840$

b ${}^{10}P_5 = 30\,240$

c ${}^9P_6 = 60\,480$

d ${}^8P_3 = 336$

e ${}^{20}P_4 = 116\,280$

f ${}^7P_2 = 42$

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